

BRIEF REPORTS

Adult Age Differences in Event-Based Prospective Memory: A Meta-Analysis on the Role of Focal Versus Nonfocal Cues

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Studies of age differences in event-based prospective memory indicate wide variation in the magnitude of age effects. One explanation derived from the multiprocess framework proposes that age differences depend on whether the cue to carry out a prospective intention is focal to ongoing task processing. A meta-analysis of 117 effect sizes from 4,709 participants provided evidence for this view, as age effects were greater when the prospective cue to the ongoing task was nonfocal compared with when it was focal. However, the results only support a weaker but not a stronger prediction of the multiprocess framework, as age impairments were reliably above zero for both types of retrieval cues.

Keywords: prospective memory, aging, meta-analysis

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Remembering to implement intended activities, such as remembering to take medication or attend a meeting at midday, is referred to as *prospective memory* (PM; Brandimonte, Einstein, & McDaniel, 1996; Kliegel, McDaniel, & Einstein, in press). PM tasks are termed *event based* when the trigger to carry out an intended action is an externally presented cue (e.g., the appearance of a mailbox). By contrast, in *time-based* PM tasks, a planned activity has to be performed at a specific point in time (e.g., at noon) or after a specified period of time (e.g., after 10 min). In the present study, we focus on event-based PM.

Developmental psychologists have investigated the question of whether the ability to accurately carry out PM tasks declines in old age. In their meta-analytic review, Henry, MacLeod, Phillips, and Crawford (2004) concluded that older adults generally perform worse than younger adults do in laboratory-based PM tasks, with particularly pronounced age deficits in event-based tasks imposing high levels of controlled strategic demands (for reviews, see also Kliegel & Jäger, 2006b; McDaniel, Einstein, & Rendell, in press). However, age differences in event-based PM in laboratory settings differ substantially across studies. Although some investigators

found substantial age-related declines in PM performance (e.g., d'Ydewalle, Luwel, & Brunfaut, 1999; Mäntylä, 1994; Maylor, 1996; Park, Hertzog, Kidder, Morrell, & Mayhorn, 1997; West & Craik, 2001), other investigators revealed that older adults perform as well as their younger counterparts do in some event-based PM tasks (e.g., Bastin & Meulemans, 2002; Einstein, Holland, McDaniel, & Guynn, 1992; Einstein & McDaniel, 1990; Einstein, McDaniel, Richardson, Guynn, & Cunfer, 1995; Jäger & Kliegel, in press; McDaniel, Einstein, Edwards, Hagoood, & Rendell, 2006; Reese & Cherry, 2002; Vogels, Dekker, Brouwer, & de Jong, 2002; West & Bowry, 2005).

In consequence, attempts have been undertaken to solve this “puzzle of inconsistent age-related declines in PM” (McDaniel et al., in press). In their multiprocess framework of event-based PM, Einstein and McDaniel argued that one important factor influencing age effects in PM may be the *focality* of PM cues (Einstein & McDaniel, 2005; McDaniel & Einstein, 2000, 2007; McDaniel et al., in press). *Focal* PM tasks are those in which the ongoing task involves processing the defining features of the PM cues (e.g., keeping words in working memory while remembering to press a button whenever a specific word appears; Einstein & McDaniel, 1990). In this case, it is assumed that the PM cues are sufficiently processed during the ongoing task to enable involuntary, automatic retrieval of the intended action. By contrast, *nonfocal* PM tasks are those in which the PM cues are not part of the information being extracted in the service of the ongoing activity (e.g., keeping words in working memory while remembering to press a button whenever the background of the screen shows a particular pattern; Park et al., 1997). In nonfocal tasks, prospective remembering is thought to require strategic attentional resources to carry out extra monitoring for the cues to perform an intended action (see Einstein

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& McDaniel, 2005, and McDaniel et al., in press, for further illustrative examples of focal and nonfocal tasks).

It is important to note that in most studies using nonfocal tasks, ongoing task processing does not focus on the target features of the PM cues that were initially associated with an intended action, even though the PM cues are perceptually embedded within ongoing task stimuli (Einstein & McDaniel, 2005; McDaniel & Einstein, 2000). For example, an ongoing task may require processing the meaning of words, whereas the PM cue is defined as a particular pair of letters embedded in the words. This task would be classified as nonfocal because the detection of pairs of letters is not required to perform the ongoing task. Another example is a task involving the processing of the semantic meaning of colored words while remembering to respond to PM cues defined as words printed in a particular color.

In light of the particular cognitive processes involved in focal and nonfocal PM tasks, it was hypothesized that age differences should be clearly evident in nonfocal PM tasks but less so in focal PM tasks, as older adults are typically impaired in strategic, effortful, but not relatively automatic cognitive processes. However, the focal–nonfocal hypothesis has directly been tested in only a small number of studies. Rendell, McDaniel, Forbes, and Einstein (2007; see also McDaniel et al., in press) recently provided initial experimental support for the expectation that age differences may be more pronounced in nonfocal compared with focal PM tasks.

In the present study, we set out to investigate the hypothesis that age differences are more pronounced in nonfocal compared with focal PM tasks by quantitatively analyzing the entire available body of literature on adult age differences in event-based PM. To this end, we classified tasks used in previous studies as focal versus nonfocal. Thereafter, we conducted a meta-analytic review on these data to test whether age effects on nonfocal PM tasks significantly exceed those of focal PM tasks. In fact, we contrasted a weaker prediction and a stronger prediction derived from the multiprocess framework: The weak version proposes that age effects are greater for nonfocal compared with focal PM tasks; the strong version goes beyond this and posits that age differences should be evident in nonfocal but not in focal PM tasks (“whether or not there are age differences will depend on whether the PM task uses nonfocal or focal target events”; Einstein & McDaniel, 2005, p. 289). Consequently, we tested whether the stronger or the weaker prediction of the multiprocess view can better account for the data.

Method

Selection of Studies

A computer-based search involving PsycINFO, PubMed, Web of Science, and PSYNDEX was conducted using the terms *prospective memory*, *delayed intention*, *aging*, *older adults*, and *development*. We also inspected the references of five overviews on age differences in PM to find to-be-included studies (Henry et al., 2004; Kliegel & Jäger, 2006b; Kliegel, Mackinlay, & Jäger, in press; McDaniel et al., in press; Phillips, Henry, & Martin, in press). The literature search was completed in November 2006.

Studies to be included in the meta-analysis had to fulfill the following criteria:

1. The study investigated groups of younger and older

adults, and all participants were healthy and community dwelling. Only groups for which the mean age exceeded 60 years were considered older. Although for younger adults the mean age was permitted to vary between 18 and 59 years, following Henry et al. (2004), we also required the mean age of the older groups to be a minimum of 15 years higher than the mean age of the younger groups. Note that if a study only reported age range instead of mean age for a given group, we calculated mean age as the mean of the age range’s two extreme values.

2. The study applied one or more laboratory-based, event-based PM tasks and reported performance in terms of number or proportion of correct PM responses.
3. The PM task consisted of multiple (i.e., more than one) trials. Hence, studies applying single-trial PM tasks were excluded because they are associated with low reliabilities (for discussion, see Maylor, 1993, 1996) and may thus result in erratic effect sizes.
4. To avoid including highly related results reported in journal articles and book chapters, we only included studies that were prepared for submission, under consideration, accepted for publication, or published in peer-reviewed journals, whereas data reported in book chapters were not included. An exception was allowed for the study of Maylor, Darbie, Logie, Della Sala, and Smith (2002) because the authors declared that they did not intend to submit the study to a journal.
5. The considered studies were written in English or in German.

Classification of PM Tasks

The PM tasks were classified as focal or nonfocal according to the descriptions of Einstein and McDaniel (2005; McDaniel & Einstein, 2000; e.g., focal cues are “PM cues that overlap with the information constellation relevant to performing the ongoing task”; nonfocal cues are “those cues that are present in the environment but not part of the information being considered by the person”; McDaniel et al., in press). From 46 studies that met the inclusion criteria, a total of 117 study-level effects were calculated, 58 for focal and 59 for nonfocal PM tasks. The initial focal–nonfocal classifications were made by Theodor Jäger. Thereafter, independent classifications for all effects were obtained from a research assistant and compared with those of Theodor Jäger. This resulted in agreement about 104 classifications and disagreement about 13 classifications between the research assistant and Theodor Jäger (Cohen’s $\kappa = .78$, $p < .001$). The 13 effects for which there was disagreement were then classified into focal and nonfocal categories by two experts (one of which was Matthias Kliegel), and final disagreements were discussed between the authors. The final classification of each task is shown in Supplemental Table 1.

Calculation of Effect Sizes

Effect sizes were calculated in terms of Hedges’s g and then transformed to unbiased estimates Hedges’s d , because the former

measure is an overestimation of effect sizes, particularly in small samples (DeCoster, 2004; Rustenbach, 2003). Accurate calculation of effect sizes depends on the availability of key information, including number of participants, means for PM performance of each group, and their accompanying standard deviations or standard errors. If these data were not reported, we computed Hedges's g from t statistics or from F statistics with one degree of freedom in the numerator. Otherwise, for three studies (i.e., Duchek, Balota, & Cortese, 2006; McDaniel et al., 2006; Zimmermann & Meier, 2006), descriptive statistics were derived from figures using the program DataThief, which recovers x and y coordinates for selected points from graphs (Version III is available at <http://www.datathief.org>; see Light, Prull, La Voie, & Healy, 2000, for an application). For one study (i.e., Salthouse, Berish, & Siedlecki, 2004), Hedges's g was derived from correlation coefficients describing the relationship between age and PM performance (DeCoster, 2004). If it was not possible to derive effect sizes from information reported in articles, authors were asked if they could provide the relevant data.¹

We avoided including dependent effect sizes in the meta-analysis to meet the assumption of statistical independence between effects. Therefore, multiple effect sizes were permitted from the same study only in cases when more than one experiment was carried out or when subgroups were created within a particular experiment so long as the groups differed from one another in terms of participants sampled (cf. Supplemental Table 1). Moreover, if a study included more than one PM performance measure for the same PM tasks (e.g., if cognitive load was manipulated within subjects) without mean PM performance across the multiple measures being reported, we calculated the arithmetic mean of multiple dependent effect sizes Hedges's g s before deriving a single effect size Hedges's d . Alternatively, we used statistics that collapsed the multiple PM performance measures, such as F statistics, for the main effect of age collapsed across a within-subjects manipulation.

Four studies reported multiple PM measures for the same samples whereby the PM measures were a mixture of tasks that could be classified as focal or nonfocal (see Supplemental Table 1). For these studies, multiple effect sizes Hedges's g s were separately averaged to a mean effect size for tasks classified as focal or nonfocal. Another exception was allowed for the study of Kidder, Park, Hertzog, and Morrell (1997) in which effect sizes were calculated for two PM tasks separately, although they were performed by the same samples, because one of the tasks included a between-subjects manipulation whereas the other did not.

Three studies reported very small differences between PM performance of younger and older adults but no standard deviations, and F values were described as being less than 1. Therefore, effect sizes could not be calculated and were thus estimated as zero for these studies (i.e., Einstein & McDaniel, 1990; Einstein et al., 1995, Experiments 2 and 3; McDaniel, Einstein, Stout, & Morgan, 2003, Experiment 2b).

Statistical Analyses

Individual effect sizes Hedges's d were pooled to derive the weighted average effect sizes d across all studies and for focal and nonfocal PM tasks separately (Hedges & Olkin, 1985; Rustenbach, 2003). Further analyses included evaluating the assumption of homogeneity of effect sizes within and between the subclasses of

focal and nonfocal tasks. Q_{wi} statistics evaluate homogeneity of effect sizes within each subclass; the sum of individual Q_{wi} s (i.e., Q_w) provides an estimate of total homogeneity. If these statistics are nonsignificant, homogeneity of effect sizes within subclasses can be assumed. Heterogeneity between the subclasses of focal and nonfocal tasks was evaluated by computing Q_B ; significant values of Q_B indicate that the partitioning into subclasses can successfully explain variance among effect sizes (Hedges & Olkin, 1985; Rustenbach, 2003).

Results

In total, data from 4,709 different (i.e., nonoverlapping) participants were incorporated (2,349 younger and 2,360 older adults). Averaged mean ages were 22.6 years (range = 18.4–59.4 years) and 70.8 years (range = 62.3–86.8 years) for the younger and older groups, respectively. Supplemental Table 1 presents all individual effect sizes for age differences in PM performance, expressed as Hedges's d . Positive values of d indicate better performance of younger adults, whereas negative values of d indicate better performance of older adults.

Weighted Average Effect Sizes

The weighted average effect size of age group across all effects was 0.64. The weighted average effect sizes were 0.54 for focal and 0.72 for nonfocal PM tasks. All these effect sizes were reliably greater than zero, $Z_s > 7.16$, $ps < .001$ (DeCoster, 2004). There were more pronounced age differences in nonfocal compared with focal PM tasks, as revealed by a Z test (cf. Henry et al., 2004), $Z = 2.20$, $p = .014$ (one-tailed p values are given because of the directional hypothesis). Excluding the five studies that violated the assumption of independence between effect sizes (see Supplemental Table 1) did not alter the pattern of results, as the age effect was still greater for nonfocal than for focal PM tasks, $Z = 2.31$, $p = .011$.

As noted above, no age differences in PM were found in three studies, but exact effect sizes could not be calculated and were thus estimated as zero. As more of these effect sizes referred to focal than to nonfocal tasks, we examined whether the zero replacement technique influenced the finding of smaller age effects for focal tasks. Specifically, we repeated the analysis after estimating effect sizes on the basis of the assumption that F values were 1 in the aforementioned studies. Because F s were reported to be less than 1 in these studies, setting F values to 1 yields upper boundaries for the effect sizes that were actually obtained and thus reflects a conservative approach in examining the hypothesis of greater age effects in nonfocal PM tasks. This analysis still revealed a larger age effect for nonfocal compared with focal PM tasks, $Z = 1.77$, $p = .038$. The result of this analysis remained the same when the five studies that violated the independence assumption were excluded, $Z = 1.91$, $p = .028$.

Tests of Homogeneity of Effect Sizes

There was homogeneity of effect sizes within each subclass: For focal PM tasks, $Q_w(57) = 36.14$, $p = .986$; for nonfocal PM tasks,

¹ We thank Elizabeth Maylor, Gilles Einstein, and Peter Rendell for providing these data.

$Q_w(58) = 42.14, p = .942$; an estimate of total homogeneity within subclasses yielded $Q_w(115) = 78.28, p = .996$. By contrast, there was heterogeneity between the subclasses of focal and non-focal tasks, $Q_B(1) = 3.06, p = .040$.

Testing for Publication Bias

To evaluate potential publication bias due to the tendency to publish significant results more often than nonsignificant results, we inspected a funnel-plot diagram plotting total sample sizes against the corresponding study-level effects (cf. Henry et al., 2004). As apparent in Figure 1, there was no relative absence of studies with small sample sizes and small effects, either for focal or for nonfocal PM tasks. Thus, there was no evidence of a potential confound of the present results due to publication bias.

Discussion

In the present study, we tested the hypothesis that deficits associated with normal aging are more pronounced in nonfocal than in focal event-based PM tasks, as nonfocal tasks are suggested to involve a higher degree of controlled, strategic cognitive processes than focal tasks (McDaniel et al., in press). As researchers in only few studies have directly examined this hypothesis (e.g., Rendell et al., 2007; cf. McDaniel et al., in press), we aimed to test the importance of cue focality in age differences in PM by conducting a meta-analysis on the body of available literature examining adult age differences in event-based PM after classifying these tasks post hoc as focal or nonfocal.

Across all studies, our analyses revealed a weighted average effect size of 0.64, which reflects a medium to large deficit of older compared with younger adults in event-based PM. Thus, old age is associated with a general decrement in performance in laboratory-based PM tasks (cf. Henry et al., 2004). This fact clearly rejects early speculations that PM may be an exception to typical age-related declines (Einstein & McDaniel, 1990; Mäntylä, 1994; but cf. Rendell & Thomson, 1999). However, comparing the present effect sizes with those found for other cognitive domains reveals that performance in event-based PM tasks is less affected by

normal aging than the recall of retrospective information, where meta-analyses have revealed greater age effects (i.e., $d = 1.01$ in Spencer & Raz, 1995; $d = 0.97$ in La Voie & Light, 1994; and $d = 0.99$ in Verhaeghen, Marcoen, & Goossens, 1993; see also Henry et al., 2004).

More important for the question being examined in this meta-analysis, the age effect was qualified by the finding that age-related declines were larger for nonfocal (i.e., $d = 0.72$) compared with focal (i.e., $d = 0.54$) PM tasks. This result is consistent with the proposal of the multiprocess framework of event-based PM (McDaniel & Einstein, 2000) that adult age differences should be greater in nonfocal compared with focal PM tasks, because controlled, strategic cognitive processes have to be engaged to a higher degree in nonfocal tasks to screen the environment for PM cues (McDaniel et al., in press; Rendell et al., 2007). However, results revealed reliable age deficits not only in nonfocal PM tasks but also in PM tasks using focal cues. This finding has implications for the multiprocess framework by indicating that the framework's stronger prediction has to be rejected, namely, the expectation that age-related declines would not be expected in focal PM tasks because such tasks can be solved by spontaneous (i.e., relatively automatic) retrieval processes. The finding of above-zero age effects in focal PM tasks thus challenges the multiprocess framework's proposal that focal tasks predominantly involve automatic retrieval processes. In contrast, the present data are clearly consistent with the weaker prediction of the multiprocess framework, as age deficits were greater for nonfocal compared with focal PM tasks, even though both types of tasks were associated with significant age-related declines. As a caveat to this conclusion, it has to be noted that most of the studies examined in this meta-analysis were not explicitly designed to directly test the impact of cue focality on age differences in PM.

Although our findings support the multiprocess framework's (weaker) prediction that older adults are more strongly impaired in PM tasks using nonfocal rather than focal cues, an alternative explanation could be derived from the fact that older adults have a reduced useful field of view (e.g., Ball, Beard, Roenker, Miller, & Griggs, 1988).² Specifically, assuming that PM cues are peripheral to (rather than embedded within) the ongoing task stimuli more frequently in nonfocal than in focal PM tasks, the possibility exists that the spatial location of PM cues may be a mediator of age differences in event-based PM rather than the focal–nonfocal distinction per se. In an attempt to examine this potential alternative explanation, we carried out an additional analysis following the same methodology as was used in the meta-analysis on focal and nonfocal cues, but we classified PM cues as central and peripheral. We defined central PM cues as cues that were perceptually embedded within ongoing task stimuli. By contrast, we defined peripheral cues as (a) cues that were presented in the spatial periphery of ongoing task stimuli or (b) cues for the detection of which a reduced useful field of view was apparently harmful because of demands for spatial attention to a wide area (e.g., a task that involves looking over the entire computer screen to detect a PM cue; e.g., Logie, Maylor, Della Sala, & Smith, 2004).

This classification yielded 94 and 23 effect sizes for central and peripheral PM cues, respectively (see the note of Supplemental Table 1 for a complete list of tasks using central or peripheral

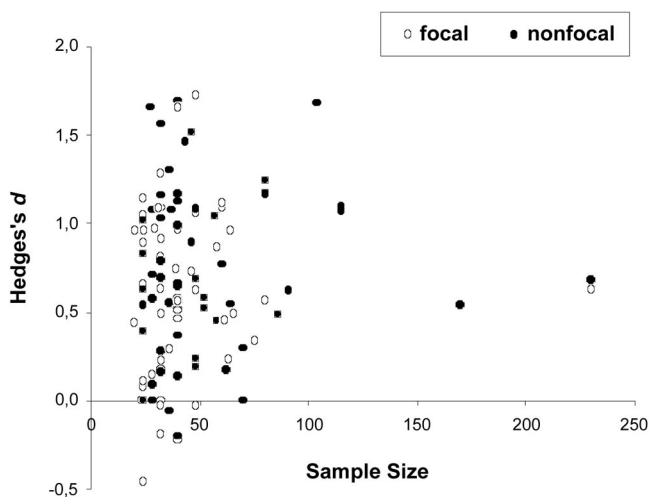


Figure 1. Study-level effects Hedges's d s plotted against total sample sizes.

² We thank an anonymous reviewer for suggesting this possibility.

cues). It is important to note that the analysis revealed that peripheral PM cues did not produce significantly larger age effects than did central PM cues (for central cues, $d' = 0.63$; for peripheral cues, $d' = 0.66$; $Z = 0.52$, $p = .300$). In conclusion, the greater age effect for nonfocal compared with focal PM tasks cannot be explained by the appearance of cues in peripheral vision. It is also worth noting that most nonfocal PM tasks actually used cues that were perceptually embedded within ongoing task stimuli (i.e., central cues).

Although cue focality appears to be one of the factors that explain inconsistencies between studies of age differences in event-based PM, the difference between age effects of focal and nonfocal tasks was only of moderate size (i.e., 0.54 vs. 0.72, respectively). In consequence, other factors besides cue focality may additionally account for the inconsistent findings across studies. For instance, tasks that are classified as focal (or nonfocal) can vary in the level of strategic demands, influencing the magnitude of age differences in PM performance as well (see Henry et al., 2004). Other factors potentially mediating age differences in PM could be characteristics of PM cues besides their focality (e.g., their familiarity or distinctiveness or the strength of the association between PM cues and intended actions), the complexity or demands of the ongoing task, the importance associated with performing a PM task, or the involvement of planning processes (see McDaniel & Einstein, 2000, for a discussion). Consequently, researchers conducting future (meta-analytic) studies may examine whether these or other factors partly mediate older adults' deficits in prospective remembering. Taken together, although many issues in the puzzle of age-related PM still remain to be solved, the findings of the present study provide meta-analytic evidence for the view that older adults are more strongly penalized in nonfocal compared with focal PM tasks.

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